

Standard form

Worksheet 1 (Non-Calculator)

1. W18/qp12/q9

$$p = 8 \times 10^{-6}$$

$$q = 2 \times 10^{11}$$

Evaluate the following, giving your answers in standard form.

(a) $p \times q$ [1]

(b) $p \div q$ [1]

(c) $\sqrt[3]{p}$ [1]

2. S19/qp11/q11

(a) Write 0.0000845 in standard form [1]

(b)

$$p = 2.7 \times 10^{11}$$

$$q = 9 \times 10^{12}$$

Evaluate the following. Give each answer in standard form.

(i) $p + q$ [1]

(ii) $p \div q$ [1]

3. S19/qp12/q12(b)

$$P = 6 \times 10^{10}$$

$$Q = 5 \times 10^9$$

Evaluate the following. Give each answer in standard form.

(i) $P - Q$ [1]

(ii) PQ [1]

4. **W19/qp11/q13**

(a) Write the number 0.00023 in standard form. [1]

(b) Evaluate $8 \times 10^9 - 9 \times 10^8$. Give your answer in standard form. [2]

5. **W19/qp12/q11(a)**

(a) Express 4500×1000^2 in standard form. [1]

6. **S20/qp12/q14**

(a) In 2017, the population of Egypt was 97500000. Write this population in standard form. [1]

(b) The population density of a country is the number of people per square kilometer. In 2017, the population of Indonesia was 2.62×10^8 , correct to 3 significant figures. The area of Indonesia is 2×10^6 , correct to 1 significant figure. Calculate an estimate for the population density of Indonesia. [2]

7. **W20/qp11/q9**

(a) Express 0.043×100^2 in standard form. [1]

(b) Evaluate

$$\frac{1.2 \times 10^7}{2 \times 10^{-3}}$$

giving your answer in standard form.

[2]

8. **W20/qp12/q11(a,b,c)**

(a) Express 340 000 in standard form.

[1]

(b) Evaluate

$$\frac{4 \times 10^7}{8 \times 10^{21}}$$

, giving your answer in standard form.

[2]

(c) $7 \times 10^a - 3 \times 10^{a-1}$. Find k.

[1]

9. **S21/qp11/q23(a(i))**

$$a = \frac{b^2 + c}{d}$$

(a) (i) Find a when $b = 4 \times 10^2$, $c = 6 \times 10^3$ and $d = 2 \times 10^2$. Write your answer in standard form.

[3]

10. **S21/qp12/q13**

(a) Write 0.000053 in standard form.

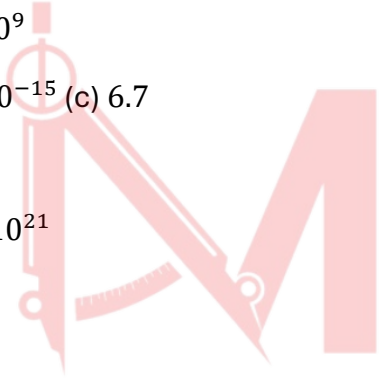
[1]

(b) Evaluate $(1.5 \times 10^{14}) \times (8 \times 10^6)$. Give your answer in standard form.

[2]

Answer Key:

1. (a) 1.6×10^6 (b) 4×10^{-17} (c) 2×10^{-2}
2. (a) 8.45×10^{-5} (b) (i) 9.27×10^{12} (ii) 3×10^{-2}
3. (b) (i) 5.5×10^{10} (ii) 3.0×10^{20}
4. (a) 2.3×10^{-4} (b) 7.1×10^9
5. 4.5×10^9
6. (a) 9.75×10^7 (b) 130 or 131
7. (a) 4.3×10^2 (b) 6×10^9
8. (a) 3.4×10^5 (b) 5×10^{-15} (c) 6.7
9. 8.3×10^2
10. (a) 3×10^{-5} (b) 1.2×10^{21}



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